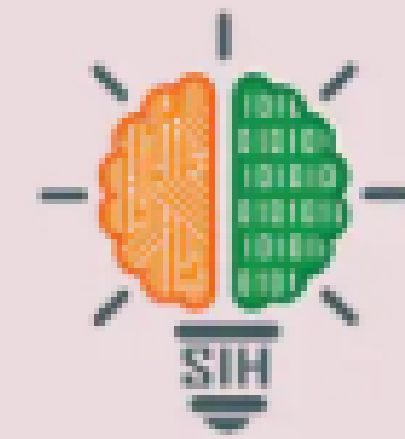
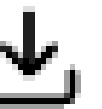


SMART INDIA HACKATHON 2024



SMART
HACKATHON
2024



TITLE PAGE

- Problem Statement ID – 1587
- Problem Statement Title – Student Innovation
- Theme – Disaster management
- PS Category – Software/Hardware
- Team ID –
- Team Name – Techbyte



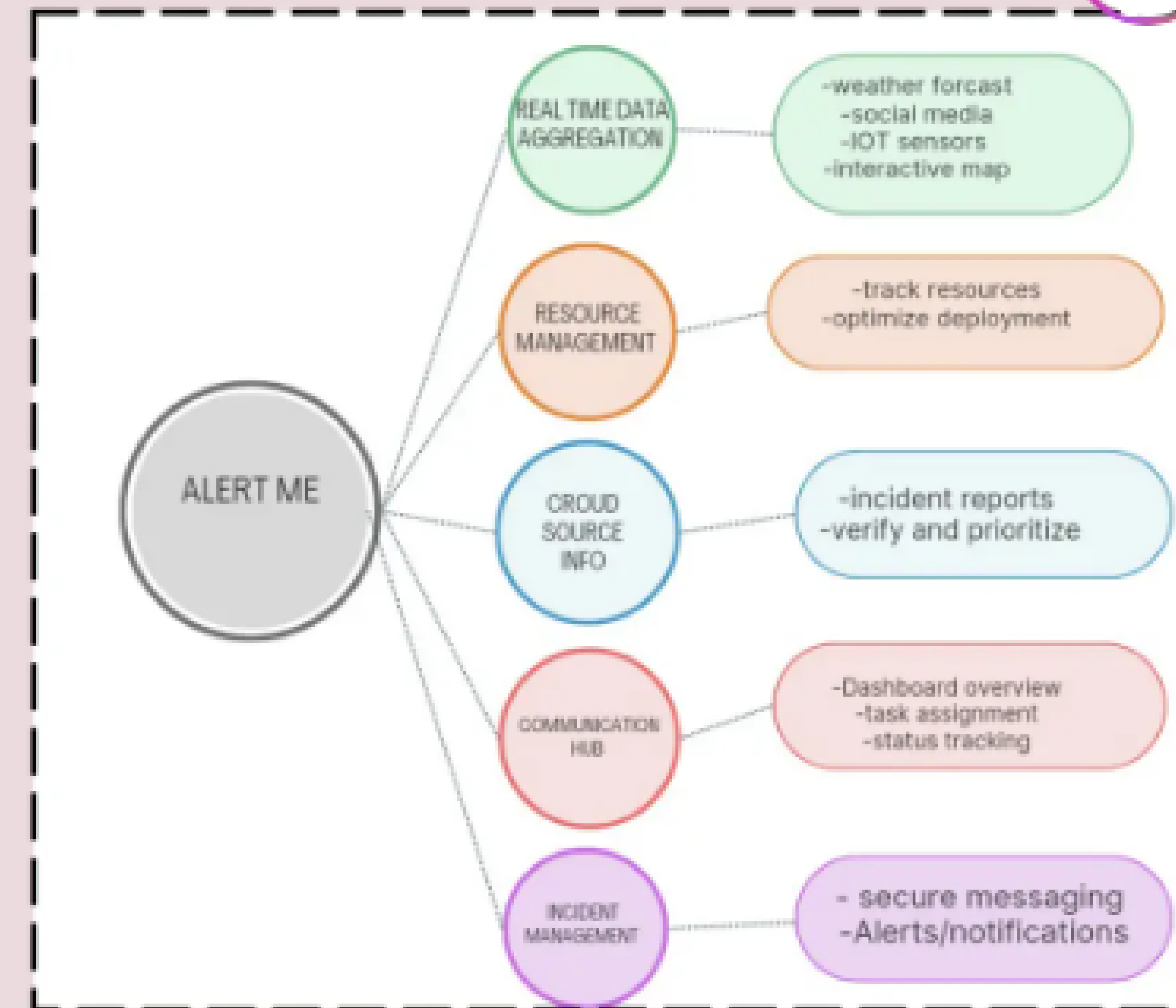
Objective: Enhance disaster response & management with real-time data and efficient coordination

OUR SOLUTION

Introducing:

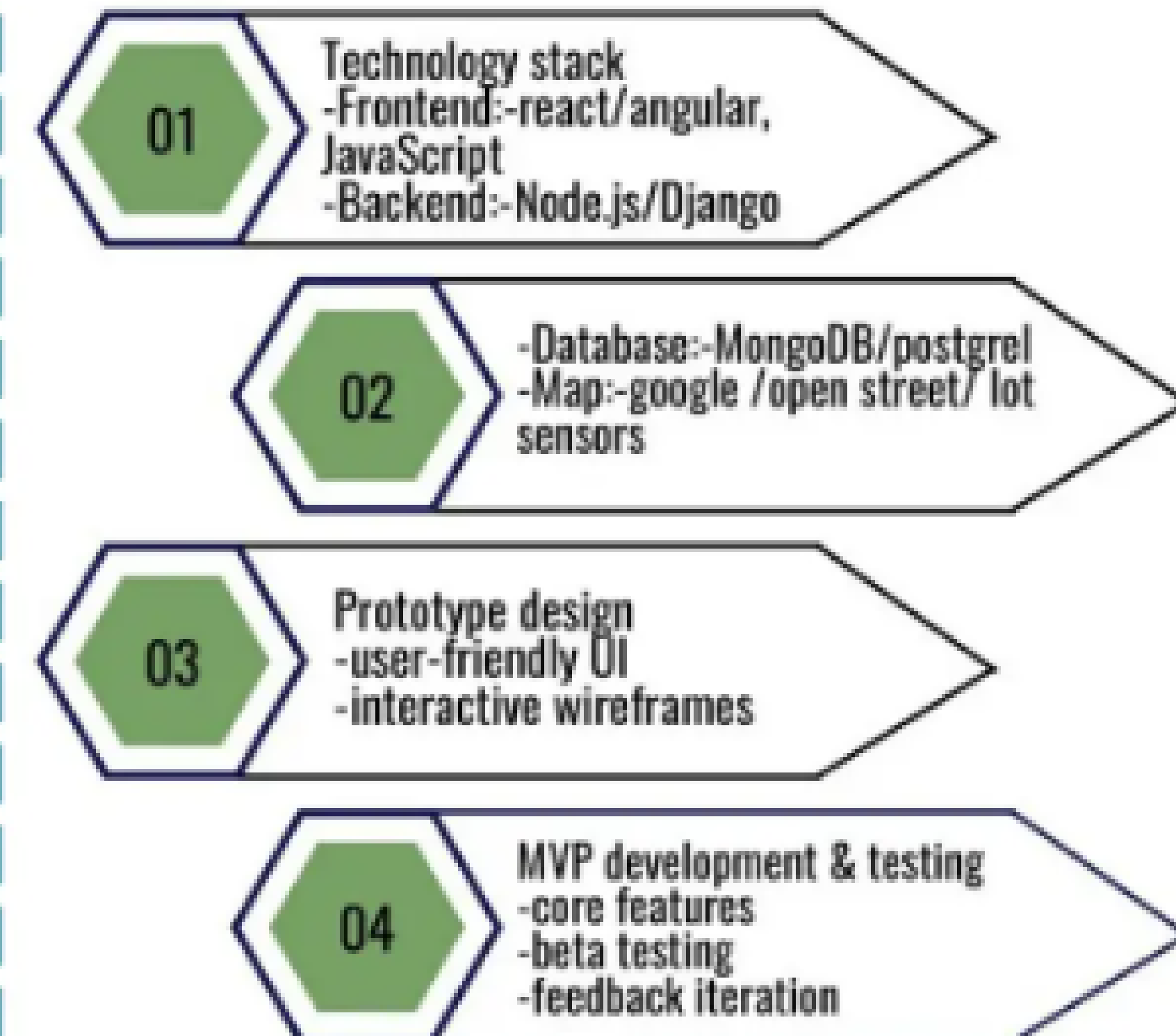
Emergency Response Coordination Platform

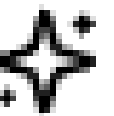
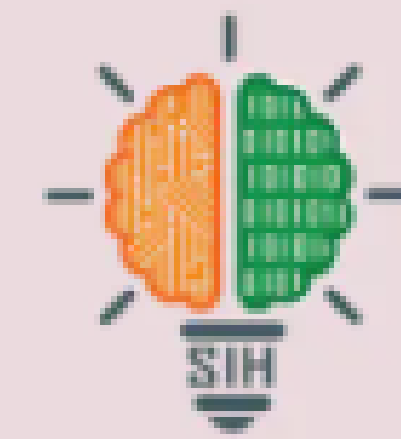
- **Real-Time Data Aggregation:** Integration of weather forecasts, social media, IoT sensors; Interactive maps for data visualization
- **Communication Hub:** Secure messaging for responders; Automated alerts for civilians
- **Resource Management:** Real-time tracking and optimization
- **Crowd-Sourced Information:** Incident reporting and verification system
- **Incident Management Dashboard:** Overview, task assignment, status tracking



TECHNICAL APPROACH

- Front-end: React/Angular (web), javascript(mobile)
- Back-end: Node.js/Django
- Database: MongoDB/PostgreSQL
- Map Integration: Google Maps/OpenStreetMap/IMD
- Cloud Services: For scalability and real-time data processing





FEASIBILITY

- **Technical Integration:** Feasible with APIs (weather, social media, IoT)
- **Technology:** React/Angular/javascript (frontend), Node.js/Django (backend), MongoDB/PostgreSQL (database)
- **Development Tools:** Modern tools and cloud services support scalability and real-time data
- **Operational:**
 - **UI:** User-friendly design achievable through iterative testing
 - **Resource Management:** Real-time tracking and optimization feasible
 - **Communication Hub:** Secure messaging and notifications available/A

VIABILITY

- **Market:**
 - Demand: High need due to increasing disasters
 - Stakeholders: Emergency teams, government, NGOs
- **Social:**
 - Impact: Enhances response efficiency and public safety
 - Accessibility: Designed for diverse user group
- **Partnerships & Funding:**
 - Partnerships: Collaboration with agencies and tech companies
 - Funding: Grants, R&D funding, and crowdfunding

IMPACT

Enhanced Response Efficiency: Real-time data & optimized resource allocation for quicker, informed decision-making

Improved Coordination: Unified communication and task management streamline response efforts

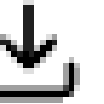
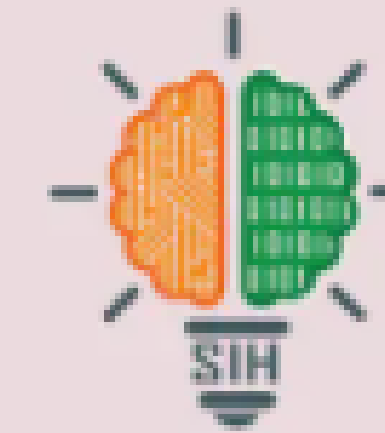
Increased Public Safety: Timely alerts & crowd-sourced information enhance situational awareness & safety

BENEFITS

Faster Response & Cost Savings: Reduces response times and minimizes resource wastage.

Better Collaboration: Centralized platform improves coordination among responders and partners.

Public Engagement: Empowers civilians to contribute and ensures accessibility for diverse user groups.



- <https://openai.com/chatgpt/>
- <https://github.com/>
- <https://www.figma.com/>
- <https://sachet.ndma.gov.in/>
- <https://www.bravostudio.app/>
- <https://www.fema.gov/emergency-managers/national-preparedness/frameworks/response>
- <https://www.gdacs.org/>